

anticlockwise to +x axis

3 B

At $t = 0$ s,

$$1.2 \times 10^5 = 2(1.0 \times 10^5) a$$

$$a = 0.60 \text{ m s}^{-2}$$

At $t = 20$ s,

$$v = at = 0.60 \times 20 = 12 \text{ m s}^{-1}$$

back carriage moves with constant speed of 12 m s^{-1} , so in another 20 s, distance moved
 $= 12 \times 20 = 240 \text{ m}$

front carriage now has acceleration of 1.2 m s^{-2} . In another 20 s,

$$s = ut + \frac{1}{2} at^2$$

$$s = 12(20) + \frac{1}{2}(1.2)(20)^2 = 480 \text{ m}$$

$$\text{distance between front and back carriage} = 480 - 240 = 240 \text{ m}$$

4 B

Ball A: